The due date for submitting this assignment has passed.
Due on 2026-03-11, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-11, 08:07 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ)

1) Which of the following statements are correct? ***1 point***

- [ ] If A and B are two square matrices, then Rank(AB)=Rank(BA).
- [x] If A and B are two 2×2 square matrices, then Rank(AB) $\leq$ Rank(B).
- [x] If A and B are two 2×2 square matrices, then Rank(AB) $\leq$ Rank(A).
- [x] If A and B are $n \times n$ matrices of rank n, then AB has rank n.

Yes, the answer is correct.
Score: 1

Accepted Answers:

If A and B are two 2×2 square matrices, then Rank(AB) $\leq$ Rank(B).
If A and B are two 2×2 square matrices, then Rank(AB) $\leq$ Rank(A).
If A and B are $n \times n$ matrices of rank n, then AB has rank n.

2) Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times3}(\mathrm{R})$) in column A with their bases in column B and the dimensions of the vector spaces in column C in Table : M2W4GA1. ***1 point***

	Vector space (Column A)		Basis (Column B)		Dimension of the vector space (Column C)
a)	$V = \left\{ \begin{bmatrix} x & y & z \\ 0 & z & x \\ y & 0 & 0 \end{bmatrix} \mid x + 2y = z, \right.$	i)	$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \right.$	1)	6